How-To Library

Informatica

Connecting to an Azure storage account using an Azure private endpoint

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Abstract

You can use an Azure private endpoint to securely and privately connect to your Azure storage account on a virtual network. This article explains how to configure an Azure private endpoint in the Azure portal.

Supported Versions

- Informatica Cloud® Data Integration Microsoft Azure Data Lake Storage Gen2 Connector

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Overview

A private endpoint is a network interface for an Azure service in your virtual network. When you create a private endpoint for your storage account, it provides secure connectivity between clients on your virtual network and your storage. The private endpoint is assigned an IP address from the IP address range of your virtual network.

Before you connect to a storage account using an Azure private endpoint, perform the following prerequisite tasks in Azure:

1. Create a network security group (optional).
2. Create a virtual network.
3. Add a subnet to the virtual network.
4. Create a virtual machine.
5. Create an Azure private endpoint.
6. Create a virtual network link.
7. Verify the Azure private endpoint configuration.

Create a network security group

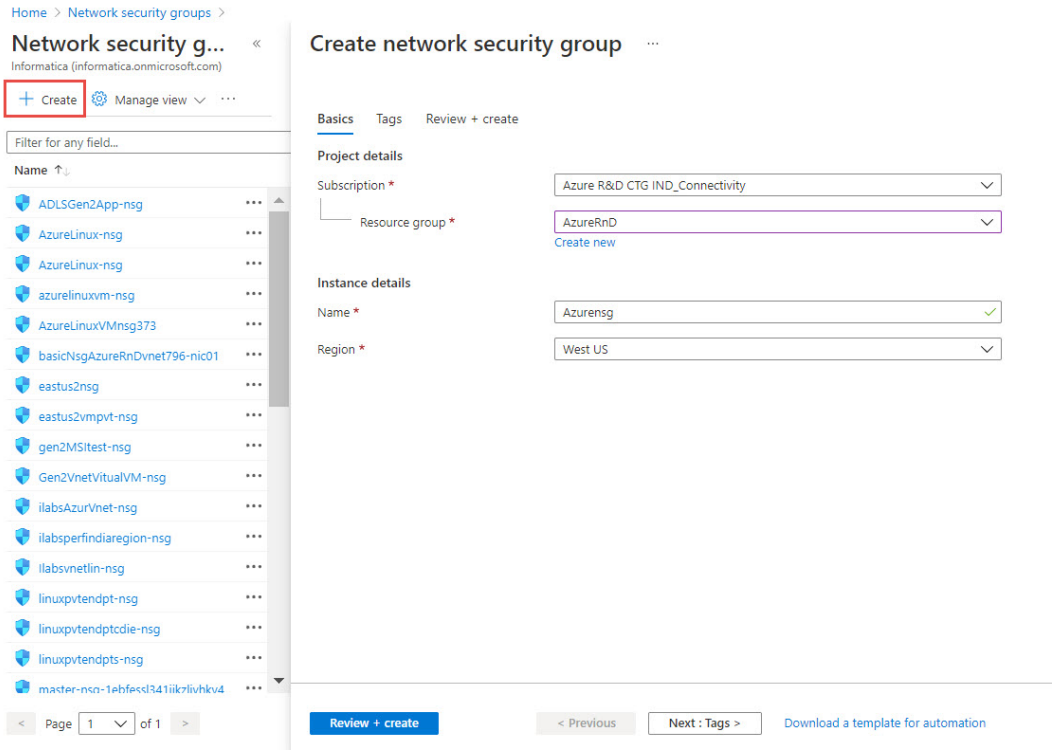
A network security group consists of a set of access control rules that allow or deny access to resources connected to Azure Virtual Networks. You can associate a network security group to subnets or individual network interfaces attached to virtual machines.

A network security group is not a mandatory requirement to use an Azure private endpoint.

Perform the following steps to create a network security group:

1. Log in to the Azure portal.

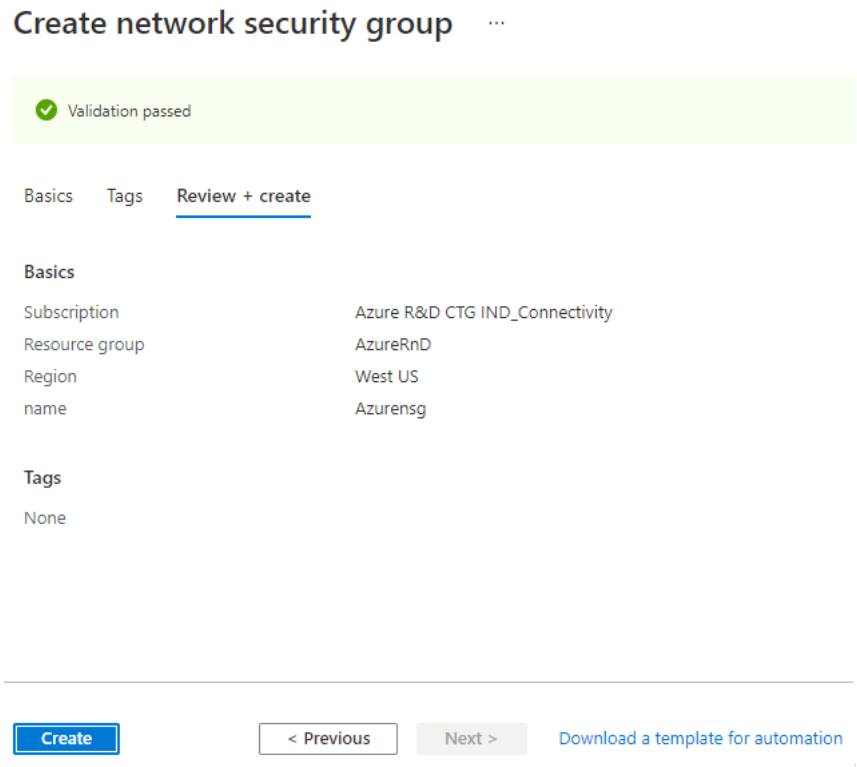
2. In the search box, enter **Network security groups** , and select **Network security groups** in the search results.
3. On the **Network security groups**, click **Create**.
4. On the **Basics** tab, enter the project and instance details.



   a. In the **Subscription** field, select your subscription for which you want to create the virtual network.
   b. In the **Resource group** field, select the resource group in which the Azure resources are deployed and managed.
   c. In the **Name** field, enter a name for the network security group.
   d. In the **Region** field, select the region.

      Note: Ensure that the network security group, the virtual network, and all the Azure resources are in the same region.

5. Click **Review + Create**, verify the configurations, and click **Create**.

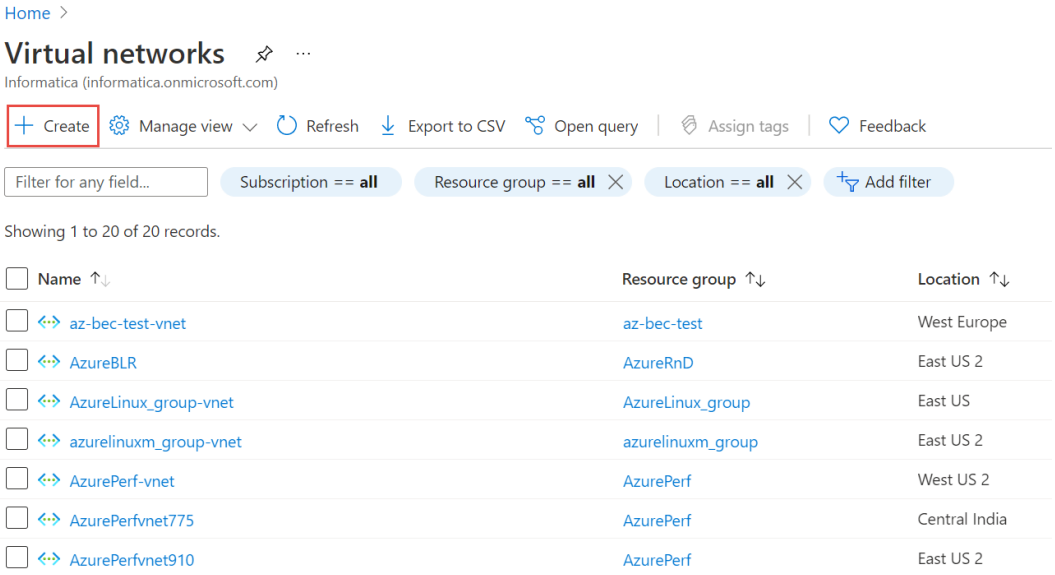


Create a virtual network

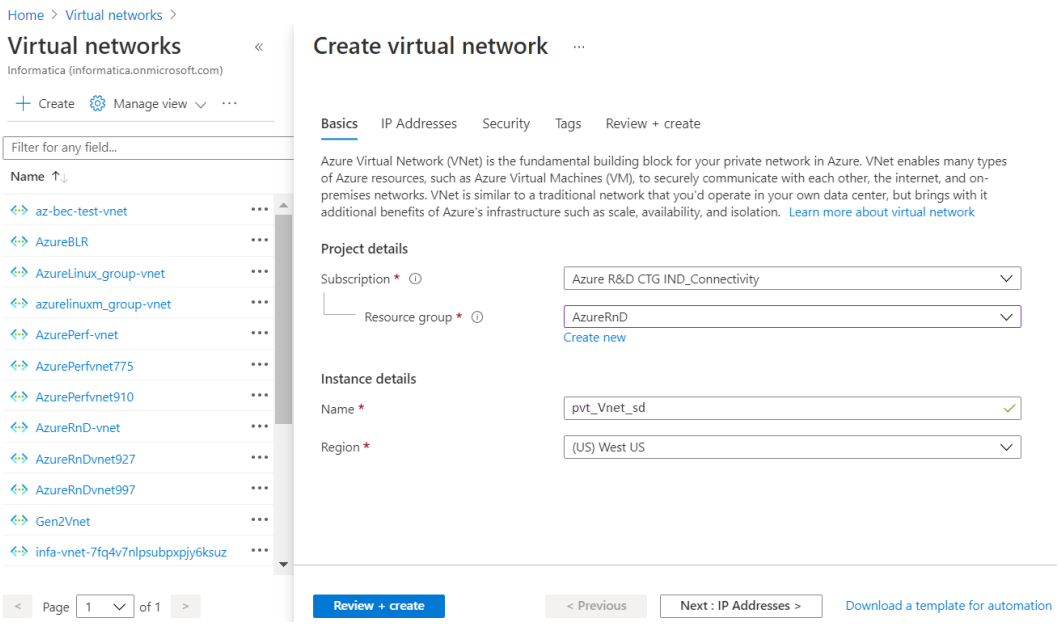
Azure Virtual Network enables Azure resources, such as Azure Virtual Machines, to securely communicate with each other, the internet, and on-premises networks.

Perform the following steps to create a virtual network in Azure:

1. In the search box, enter **Virtual networks**, and select **Virtual networks** in the search results.
2. On the **Virtual networks** page, click **Create**.



3. On the **Basics** tab, enter the project and instance details.



   a. In the **Subscription** field, select your subscription for which you want to create the virtual network.

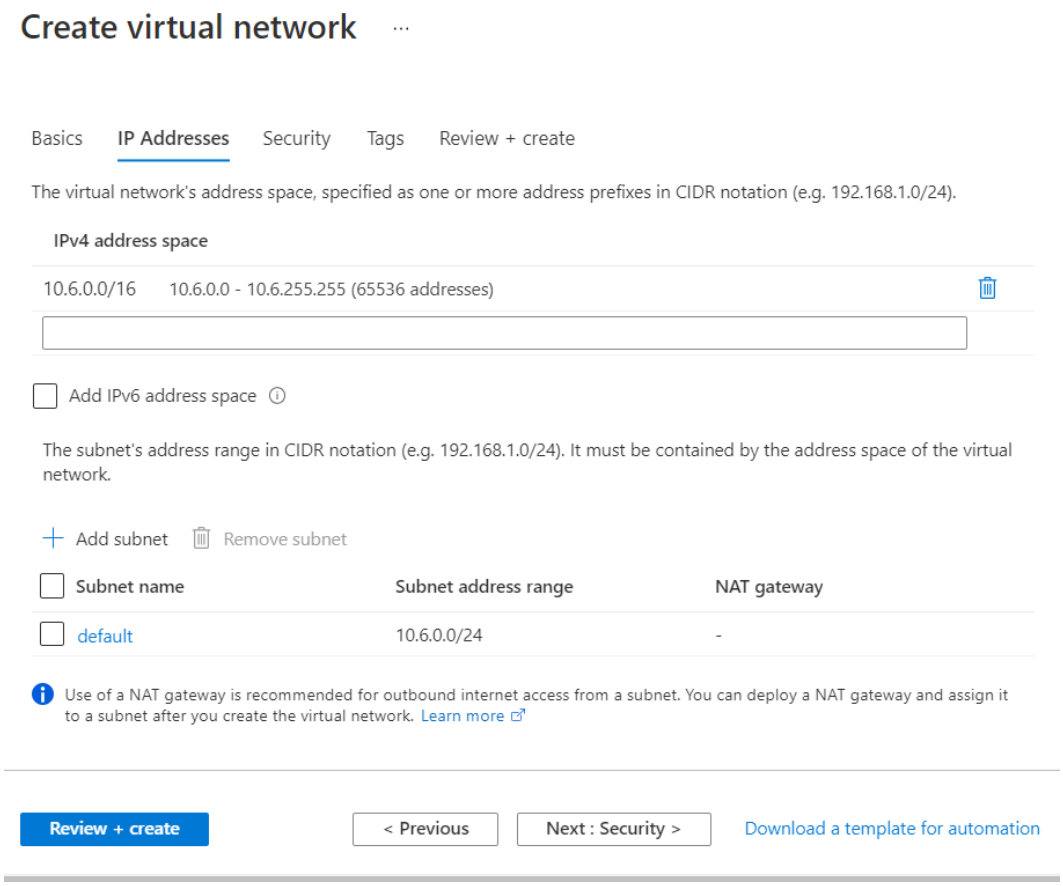
   b. In the **Resource group** field, select the resource group in which the Azure resources are deployed and managed.

   c. In the **Name** field, enter a name for the virtual network.

   d. In the **Region** field, select the region.

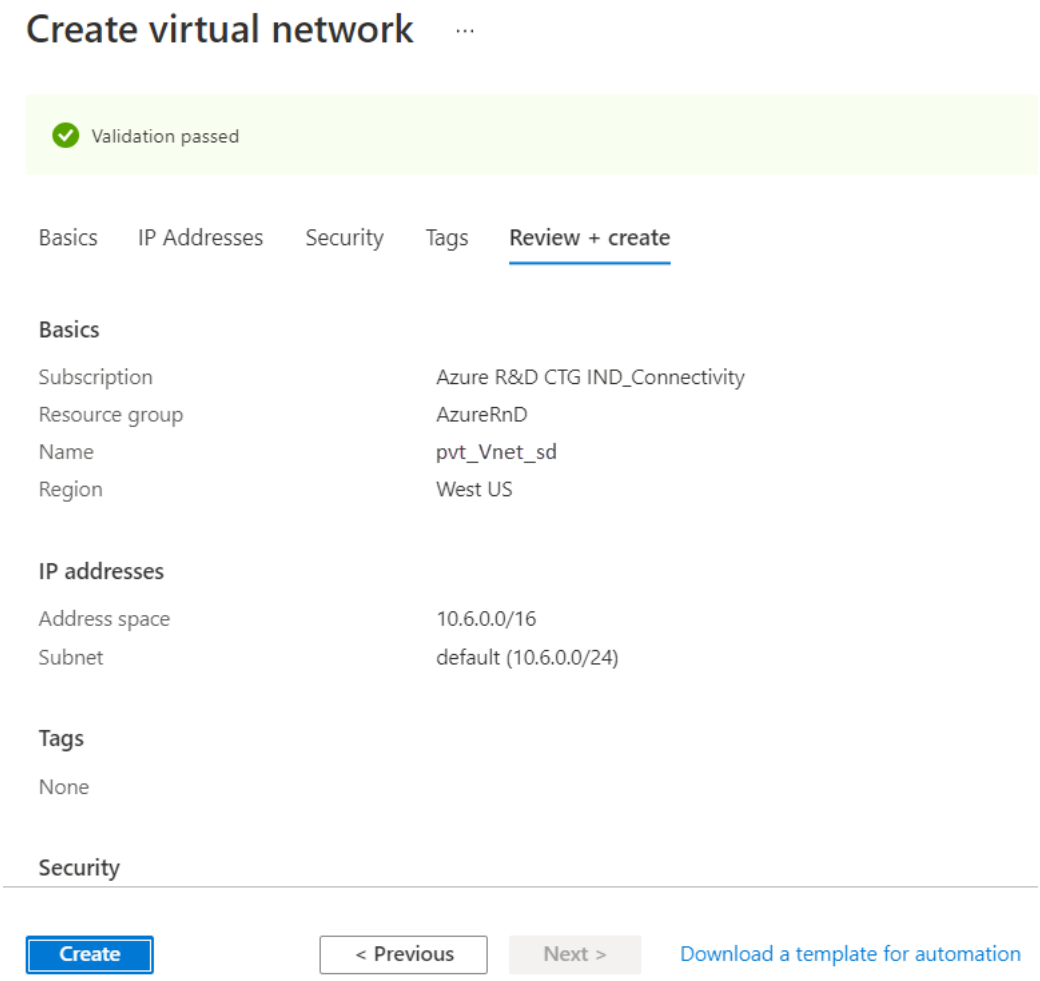
      Note: Ensure that the virtual network and all the Azure resources are in the same region.

4. Click **Next: IP Addresses**.
   The IP Addresses tab shows the IP address space of the virtual network and the address range of the subnet.



You can use the default subnet or add a new subnet. The subnet address range must be contained by the address space of the virtual network.

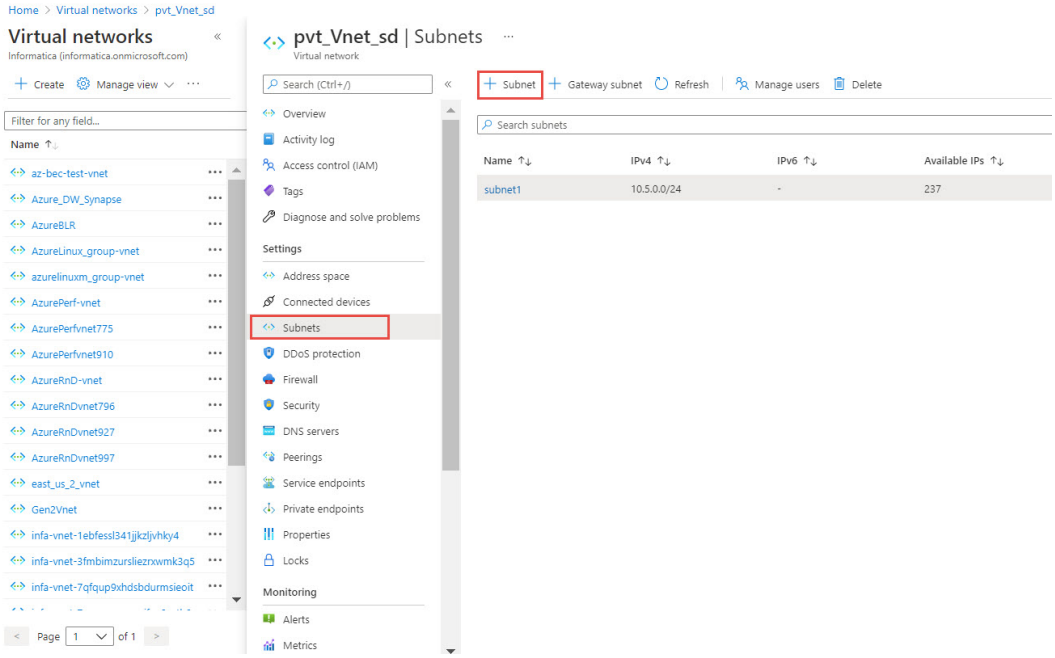
5. Click **Review + Create**, verify the configurations, and click **Create**.



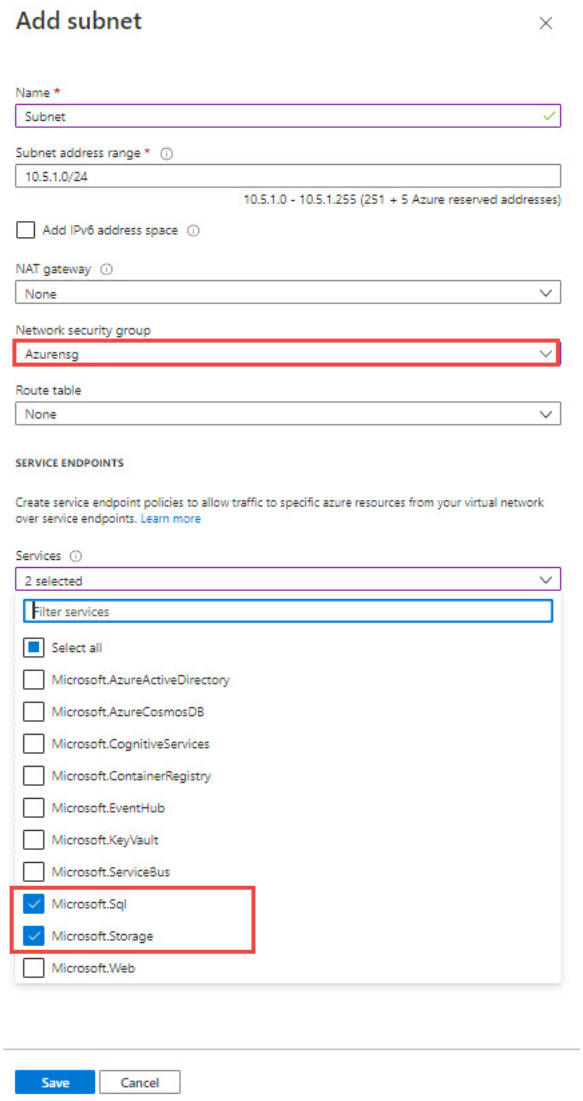
Add a subnet to the virtual network

A subnet is a range of IP addresses in the virtual network. You can segment the virtual network into one or more sub-networks and allocate a portion of the virtual network's address space to each subnet. You can then deploy Azure resources in a specific subnet.

1. Go to the virtual network that you created.
2. Under **Settings**, click **Subnets**.



3. Click **Subnet**.
   The **Add subnet** page appears.

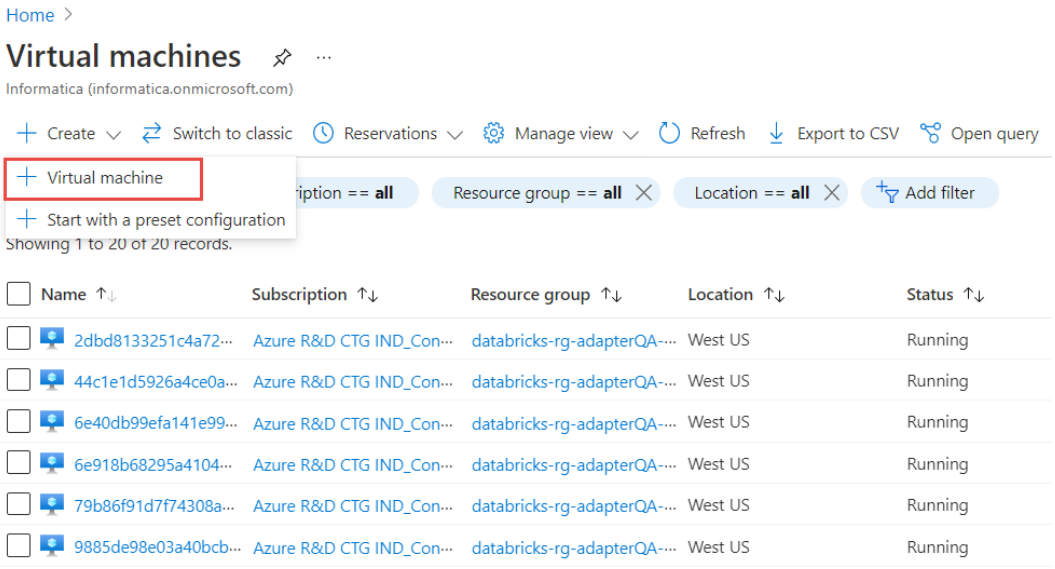


4. In the **Name** field, enter a name for the subnet.
5. In the **Subnet address range** field, you can specify an address range as per your requirement or use the default subnet address range .
   The subnet address range must be contained by the address space of the virtual network. You cannot edit the address range of a subnet which is in use.
6. In the **Network security group** field, select the network security group that you created. If you do not want to use a network security group, select **None**.
7. In the **Services** field, select **Microsoft.Sql** and **Microsoft.Storage**
8. Click **Save**.

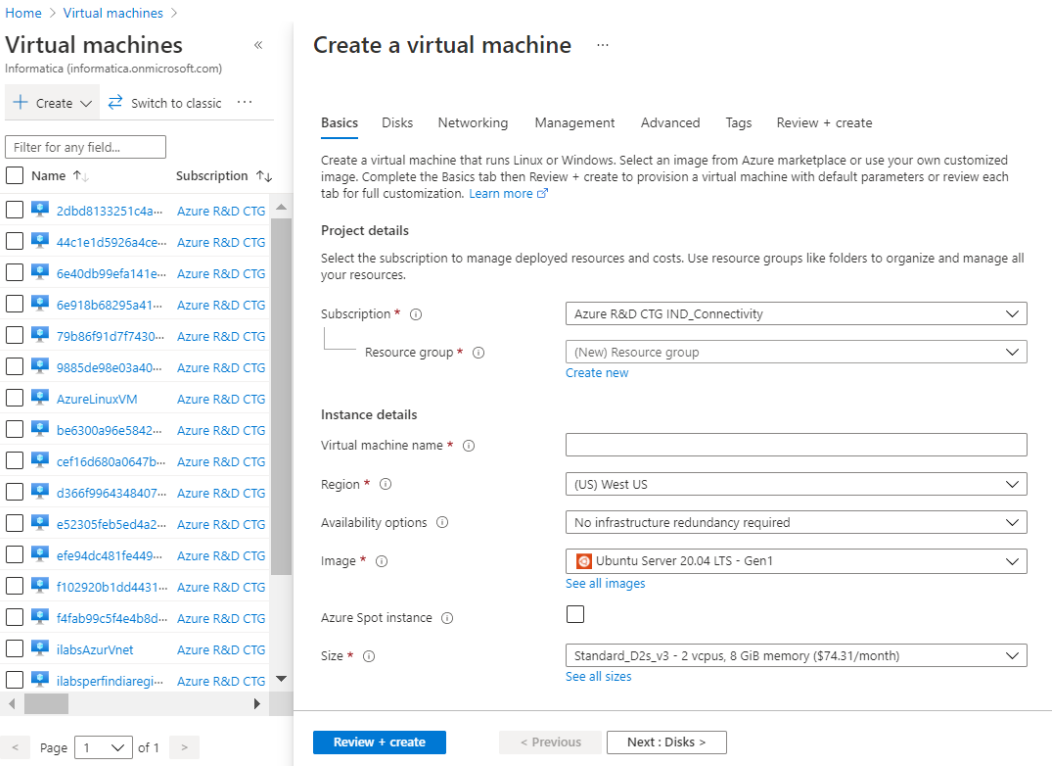
Create a virtual machine in the subnet

Perform the following steps to create a virtual machine:

1. In the search box, enter **Virtual machines** , and select **Virtual machines** in the search results.
2. Click **Create** > **Virtual machine**.



3. On the **Basics** tab, enter the project, instance, and authentication details.



a. In the **Subscription** field, select the subscription for which you want to create the virtual machine.

b. In the **Resource group** field, select the resource group in which the Azure resources are deployed and managed.

c. In the **Virtual machine name** field, enter a name for the virtual machine.
Once you create the virtual machine, you cannot change the virtual machine name.

d. In the **Region** field, select the region.

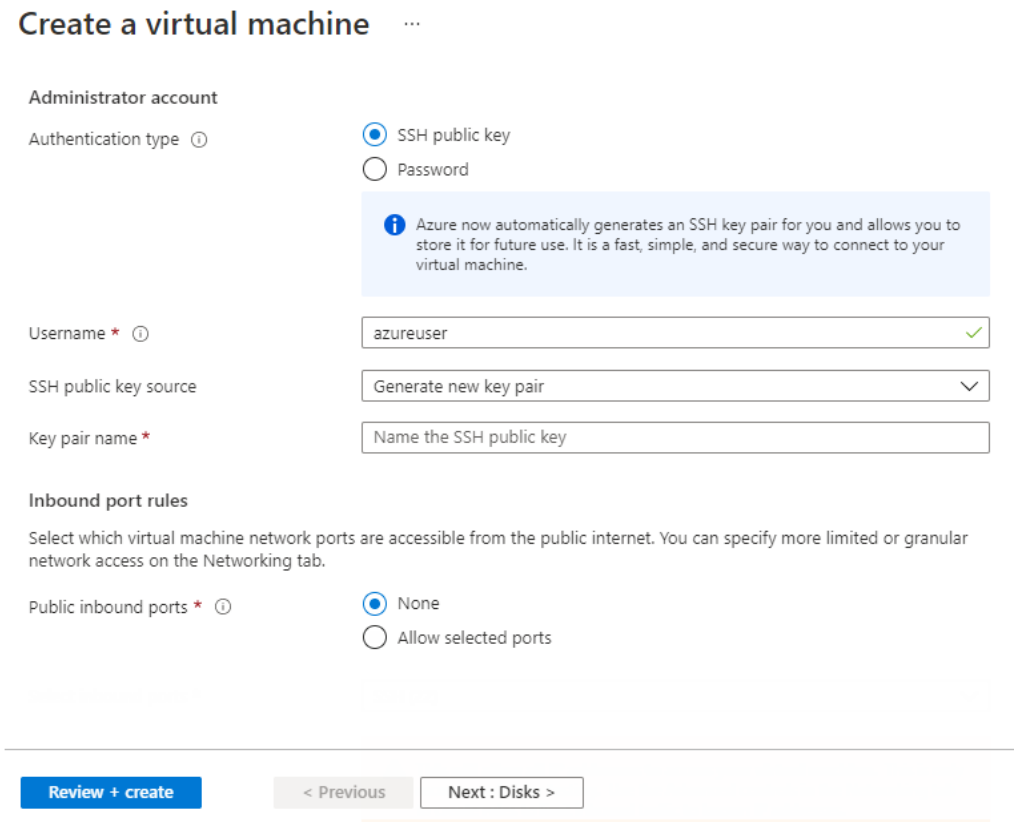
Note: Ensure that the subscription, resource group, and region for the virtual machine are the same as that of the virtual network.

e. In the **Availability** options field, you can choose to replicate the virtual machine in availability zones or availability sets to protect your applications and data from datacenter outages and maintenance events.

f. In the **Image** field, select the base operating system or application for the virtual machine.

g. In the **Size** field, select the size of the virtual machine that determines factors such as processing power, memory, and storage capacity.

h. In the **Authentication type** field, select if the administrator account must use the user name and password or SSH keys for authentication.



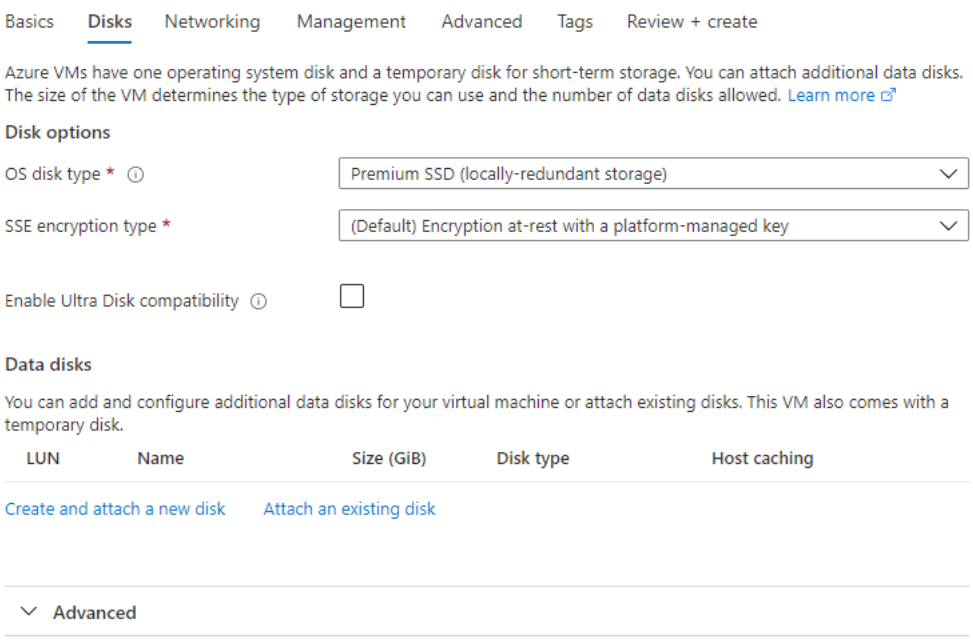
If you select the **SSH public key** option, enter the user name and key pair name.

If you select the **Password** option, enter the values in the username, password, and confirm password fields.

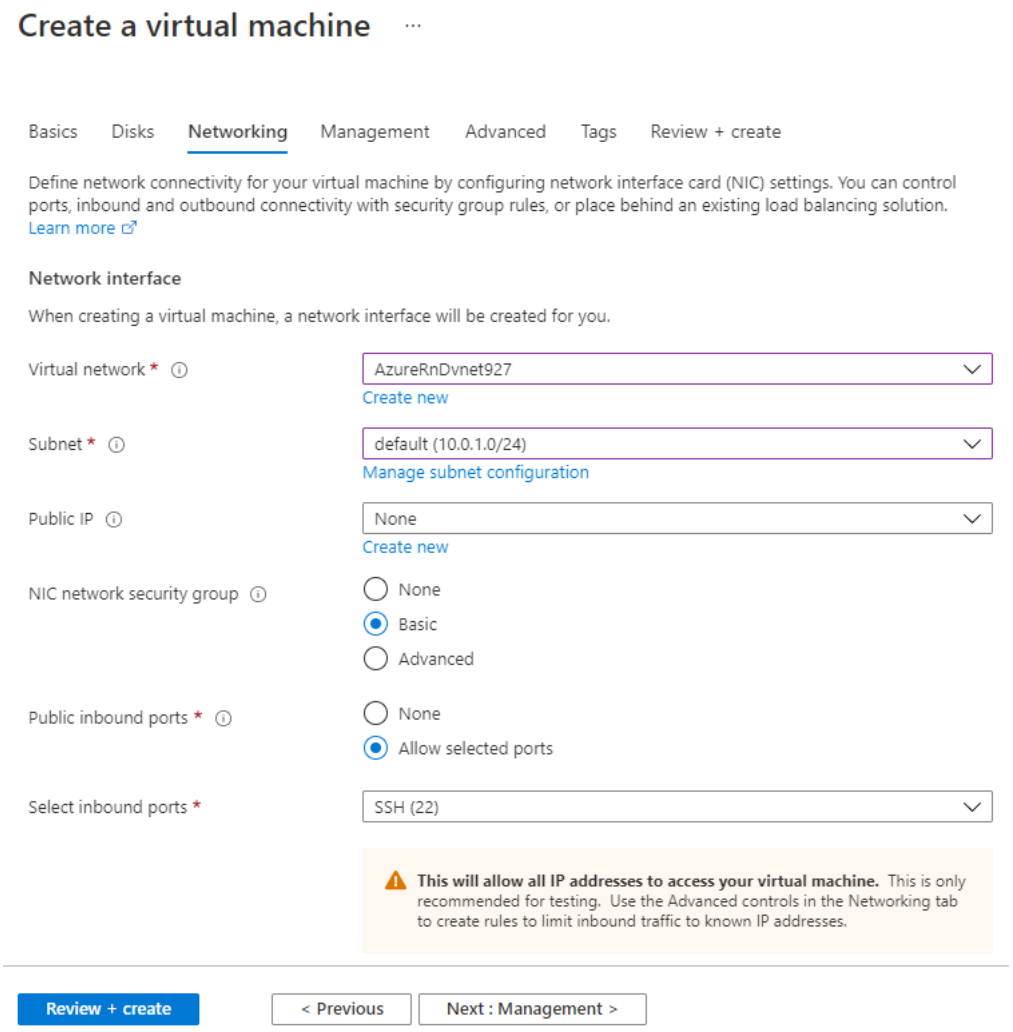
i. In the **Public inbound ports** field, select **None**.

j. Click **Next : Disks**.

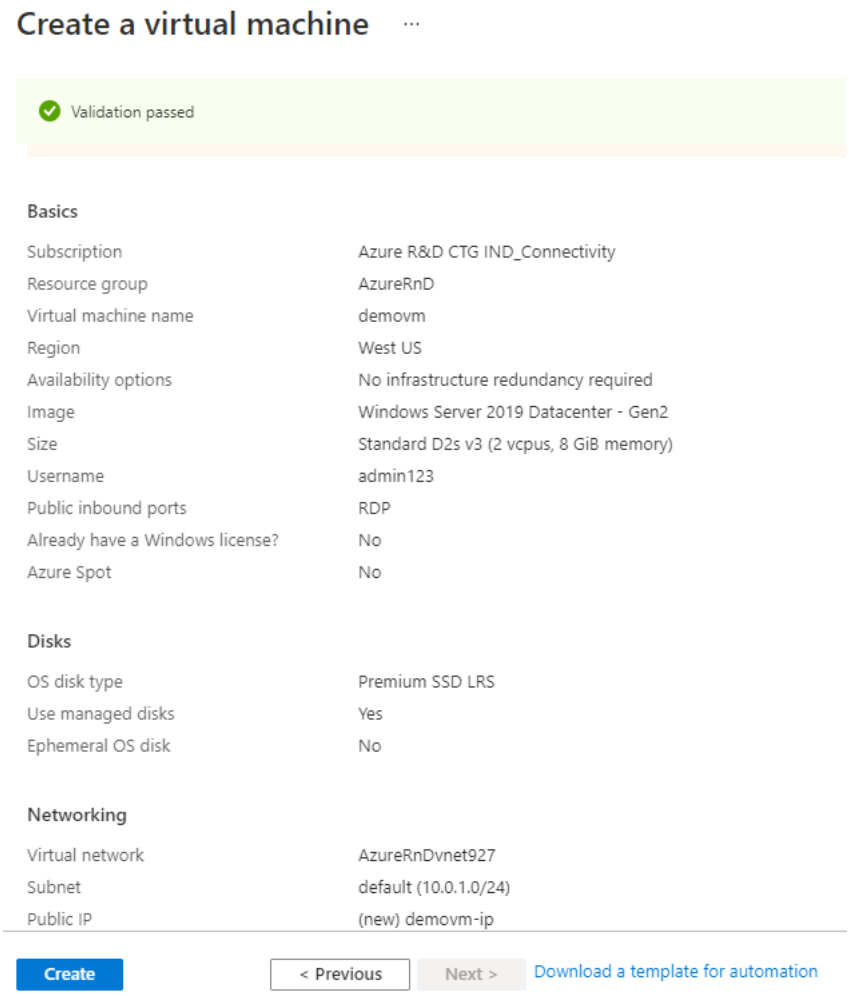
4. On the **Disks** tab, you can select the disk type for your virtual machine or use the default disk type. You can also configure additional data disks or attach existing disks.



5. Click **Next : Networking**.
6. On the **Networking** tab, select the virtual network and the subnet that you created, and then click **Review + create**.



7. On the **Review + create** tab, verify the configurations for the virtual machine.



8. Click **Create**.

Create an Azure private endpoint

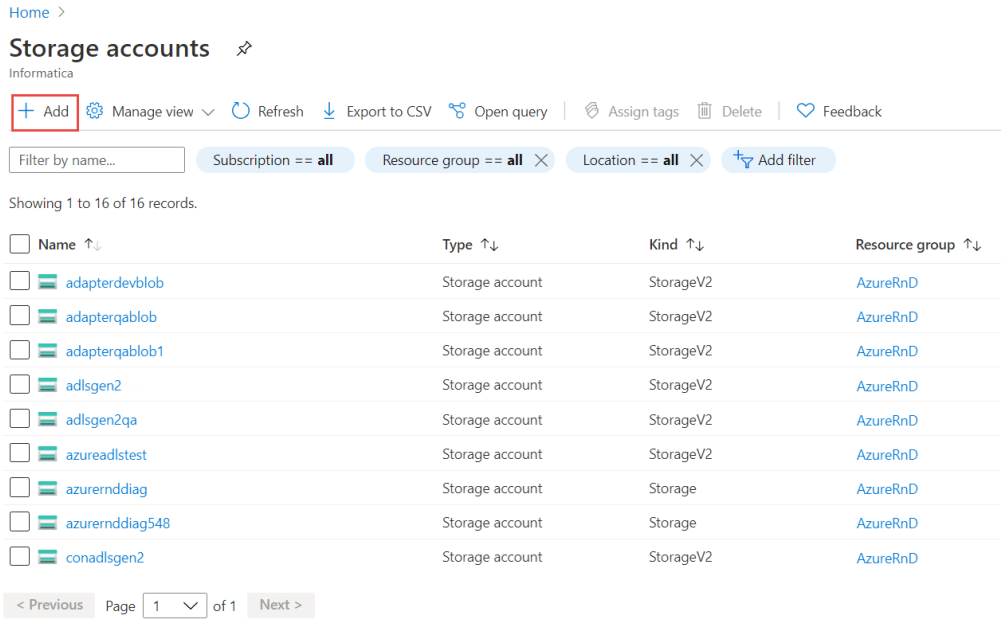
You can create a private endpoint for an existing or new storage account. You can create a private endpoint in the **Networking** tab of your storage account.

Creating a private endpoint for a new storage account

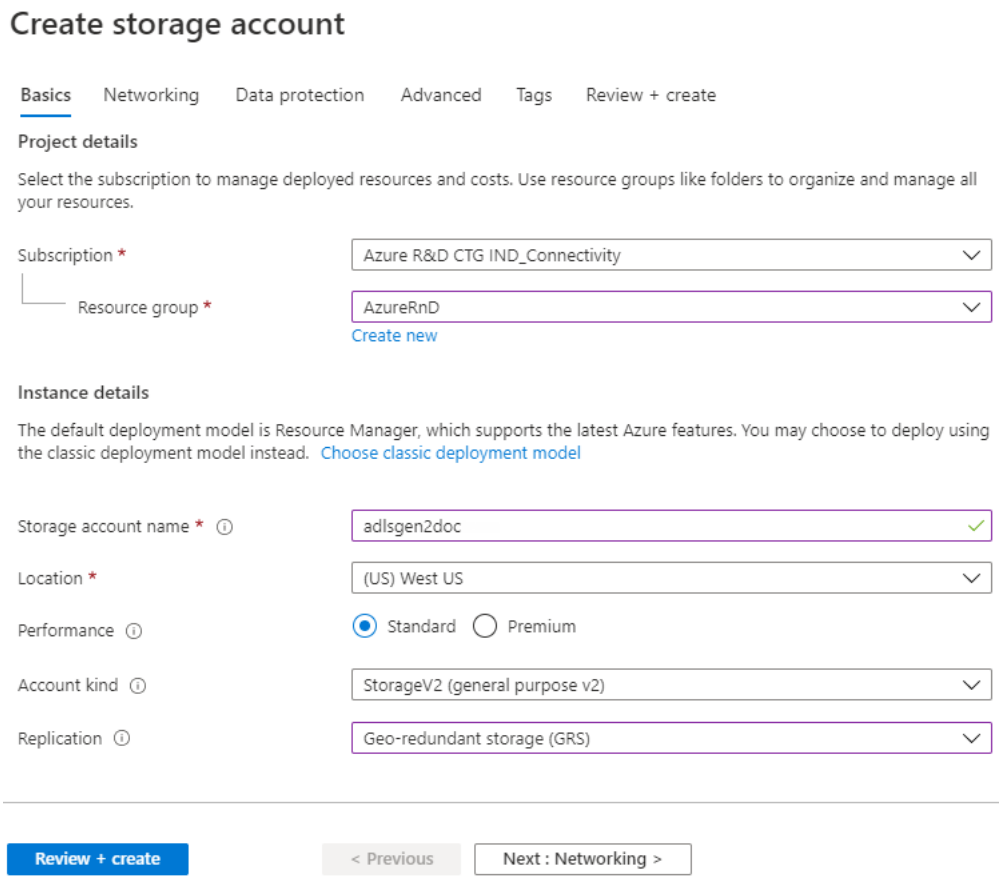
Perform the following steps to create a private endpoint for a new storage account:

1. In the search box, enter **Storage accounts**, and select **Storage accounts** in the search results.

2. On the **Storage accounts** page, click **Create** to create a new storage account.



3. On the **Basics** tab, enter the project and instance details.



a. In the **Subscription** field, select the subscription in which you want to create the storage account.

b. In the **Resource group** field, select the resource group in which the Azure resources are deployed and managed.

c. In the **Storage account name** field, enter a name for your storage account.

   Note: The name must be unique across Azure, between 3 and 24 characters in length, and must include only numbers and lowercase letters.

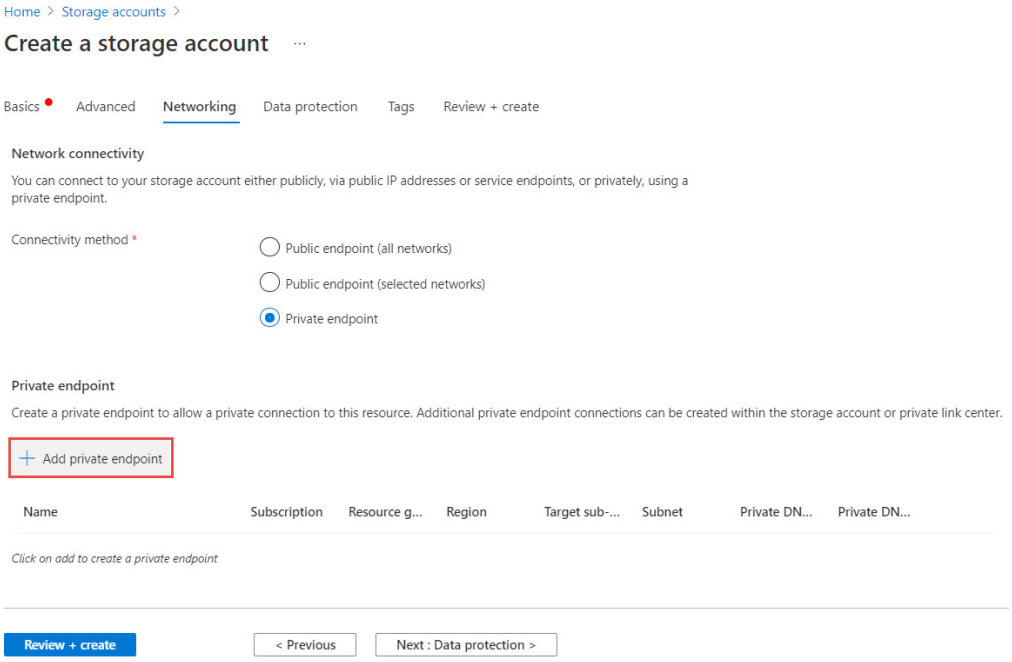
d. In the **Location** field, select a location for your storage account, or use the default location.

e. In the **Performance** field, select **Standard**.

f. In the **Account kind** field, select **StorageV2 (general purpose v2)**.
   A general-purpose v2 storage account provides access to all the Azure Storage services, such as blobs, files, queues, tables, and disks.

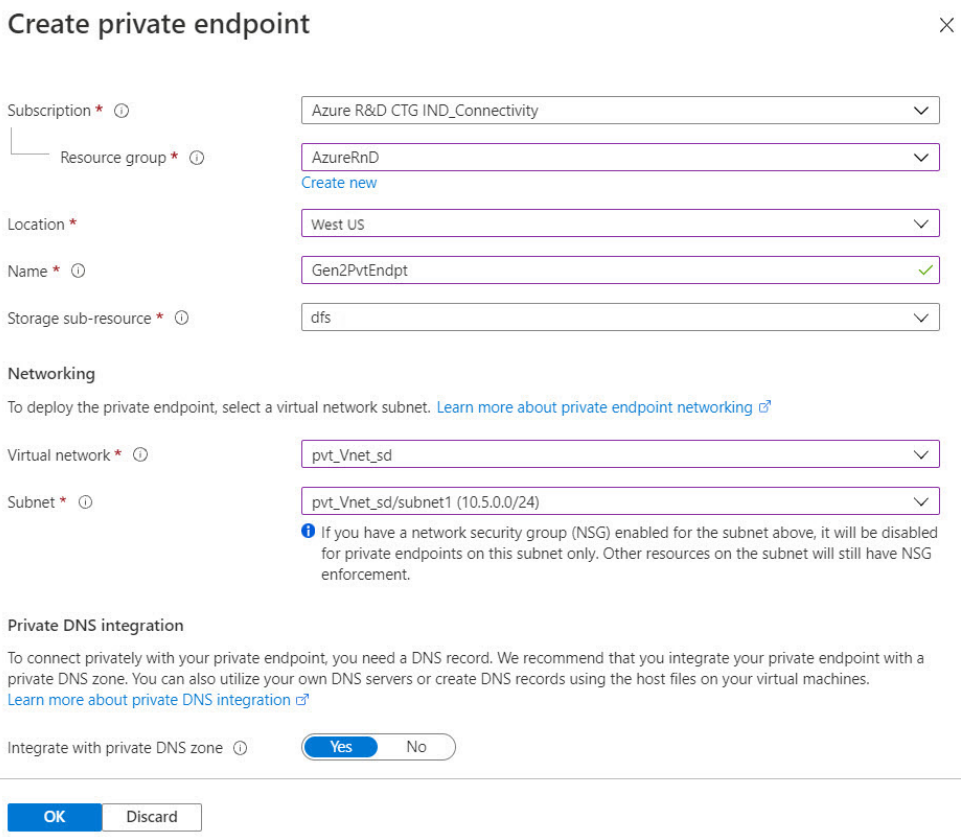
g. In the **Replication** field, select **Geo-redundant storage (GRS)**.
   The replication type specifies how the storage account will be replicated.

4. On the **Networking** tab, select the **Connectivity method** as **Private endpoint**.



a. Click **Add private endpoint**.

b. In the **Subscription** field, select the subscription for which you want to create the private endpoint.



c. In the **Resource group** field, select the resource group in which the Azure resources are deployed and managed.

d. In the **Location** field, select the location.

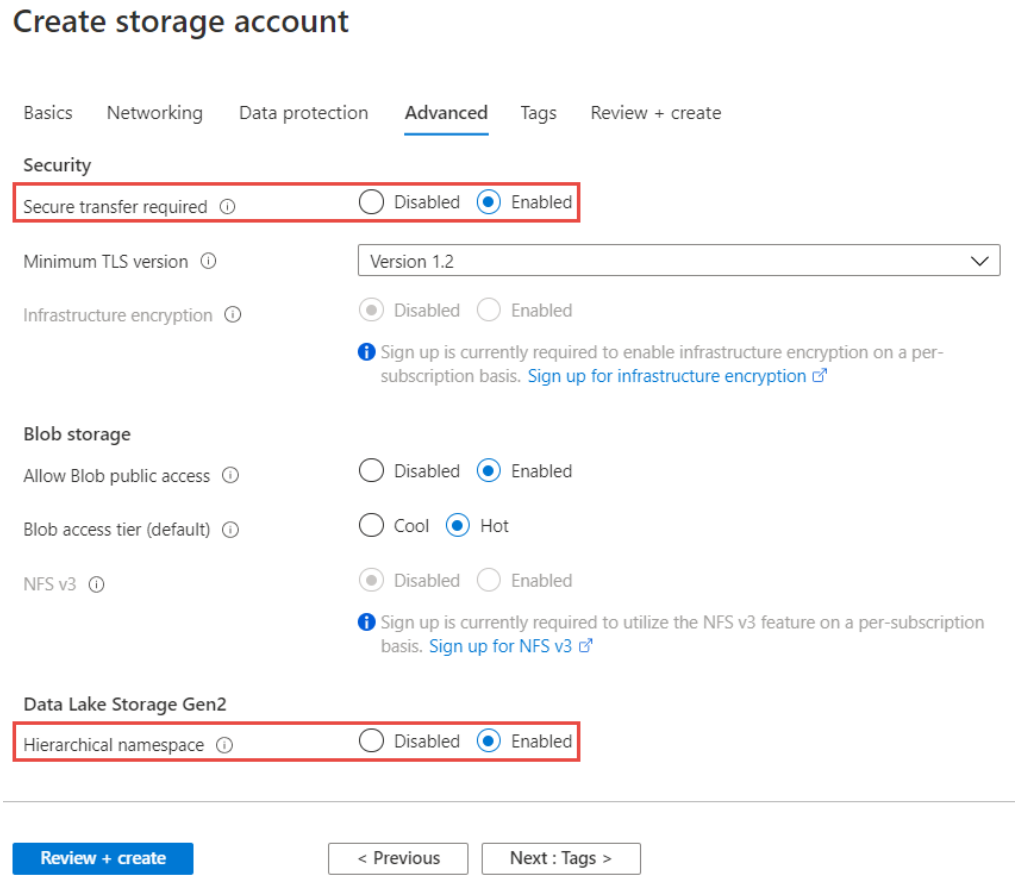
Note: Ensure that the subscription, resource group, and location for the private endpoint are the same as that of the virtual network.

e. Enter a name for the private endpoint.

f. In the **Storage sub-resource** field, select **dfs** to connect to Microsoft Azure Data Lake Storage Gen2 as the storage account.

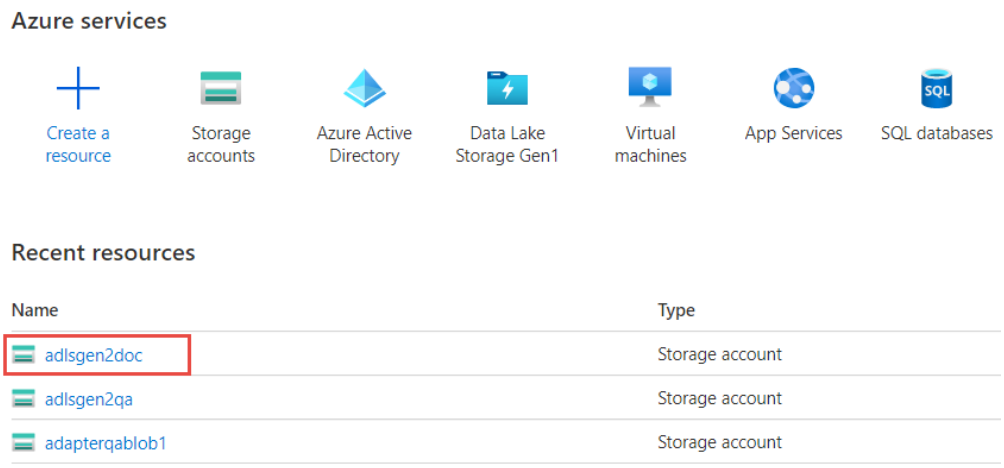
g. In the **Networking** section, select the virtual network and the subnet that you created.

5. On the **Advanced** tab, set the **Secure transfer required** and **Hierarchical namespace** fields to **Enabled**.



6. Click **Review + Create** > **Create**.

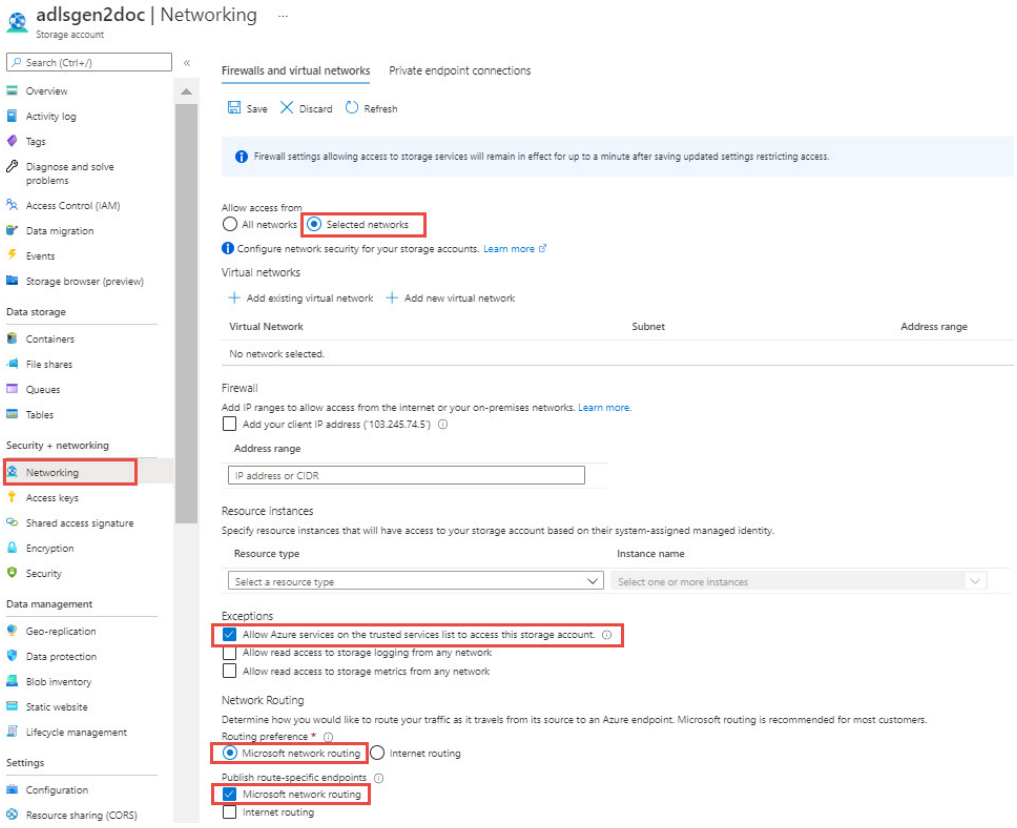
7. Click on the newly created storage account name.



Creating a private endpoint for an existing storage account

Perform the following steps to create a private endpoint for an existing storage account:

1. Navigate to the storage account for which you want to create a private endpoint.
2. Click **Networking**.



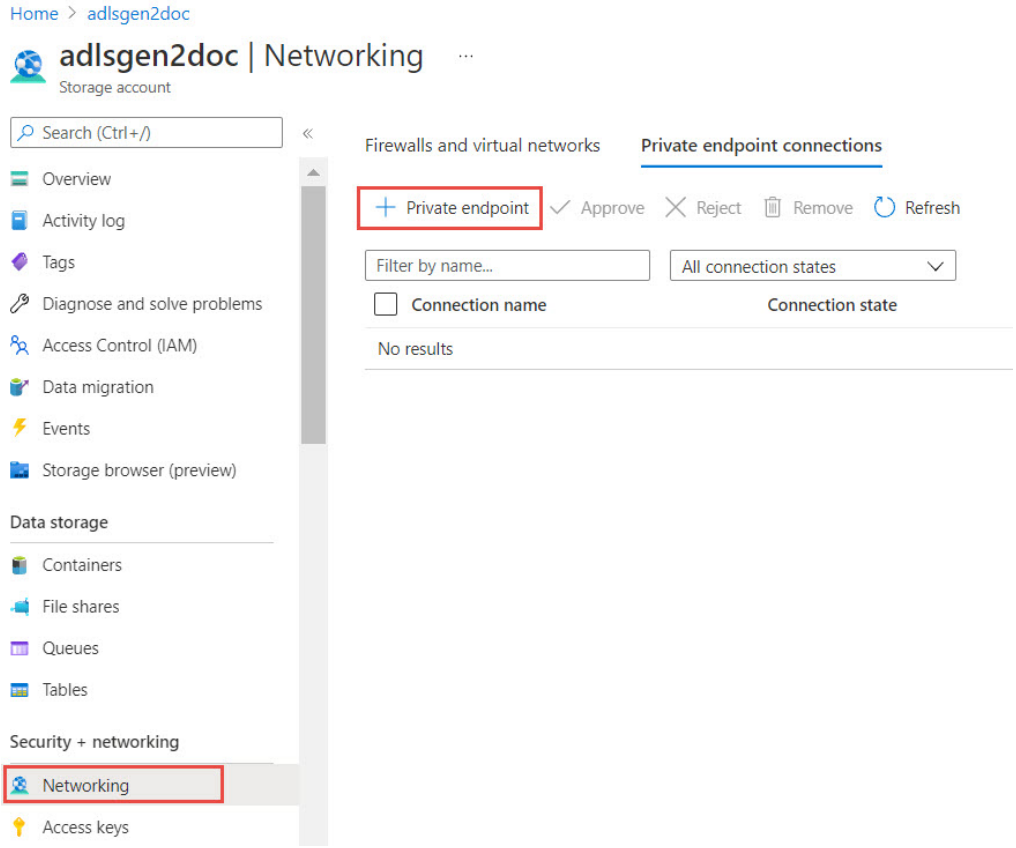
3. On the **Firewalls and virtual networks** tab, configure the following fields:
   a. In the **Allow access from** field, select **Selected networks**.

b. In the **Exceptions** section, select **Allow Azure services on the trusted services list to access this storage account**.

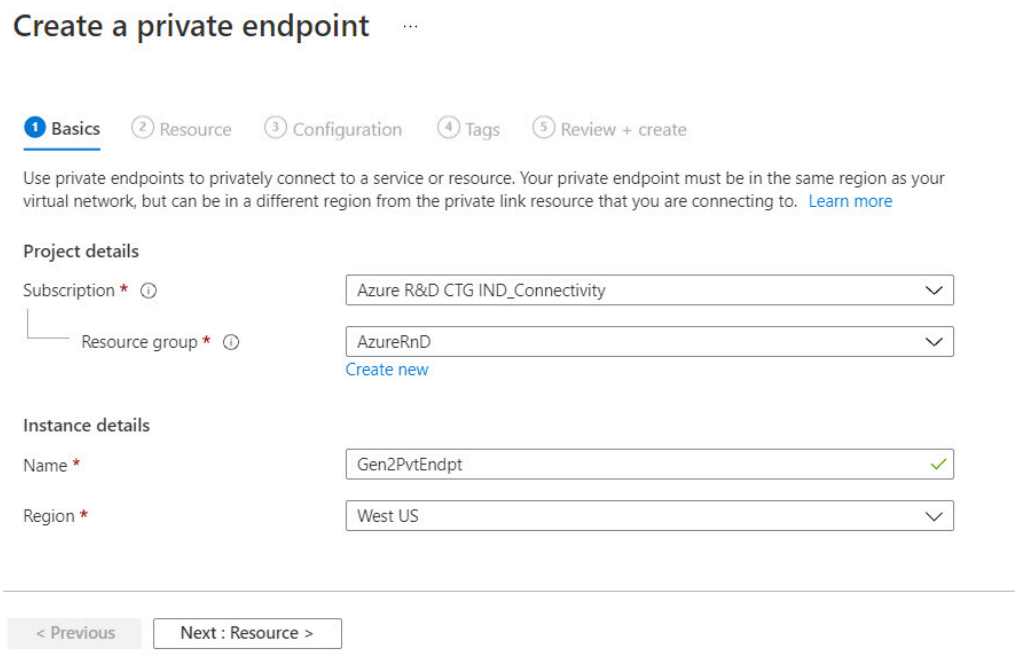
c. In the **Network Routing** section, select **Microsoft network routing**.

d. In the **Publish route-specific endpoints** section, select **Microsoft network routing**.

4. On the **Private endpoint connections** tab, click **Private endpoint**.



5. On the **Basics** tab, enter the project and instance details.



a. In the **Subscription** field, select the subscription for which you want to create the private endpoint.

b. In the **Resource group** field, select the resource group in which the Azure resources are deployed and managed.

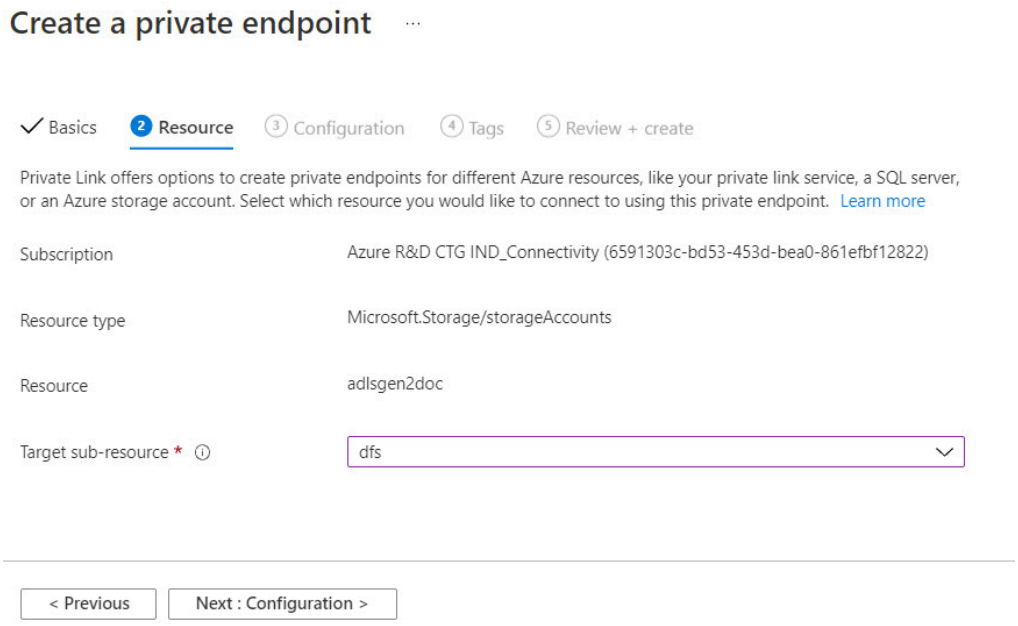
c. Enter a name for the private endpoint.

d. In the **Region** field, select the location for the private endpoint.

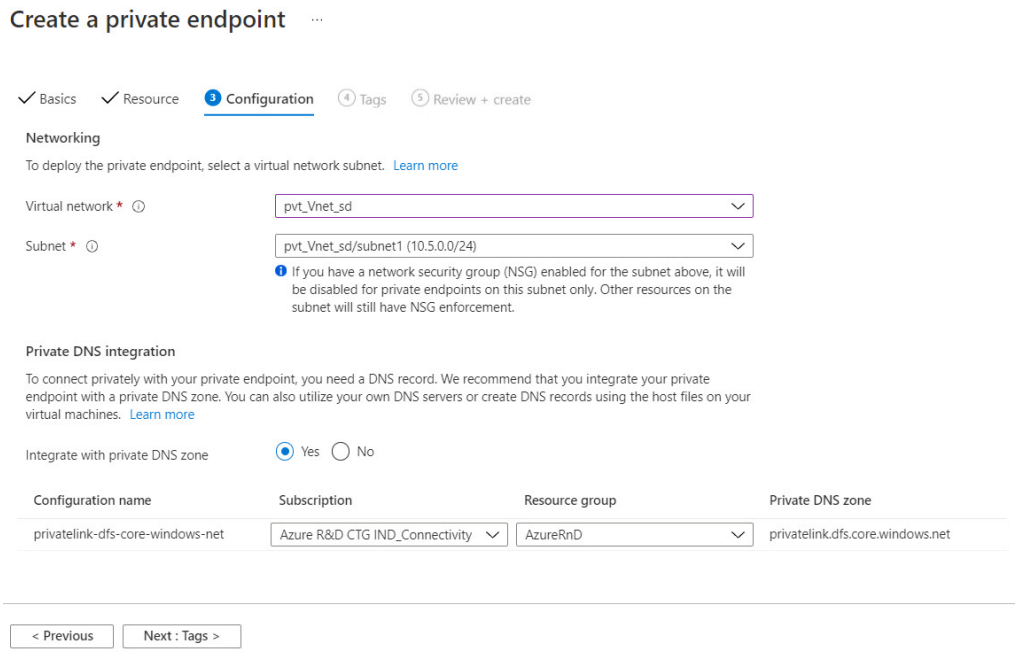
   Note: Ensure that the subscription, resource group, and location for the private endpoint are the same as that of the virtual network.

e. Click **Next : Resource**.

6. On the **Resource** tab, select the **Target sub-resource** as **dfs** to access the storage account, and then click **Next : Configuration**.

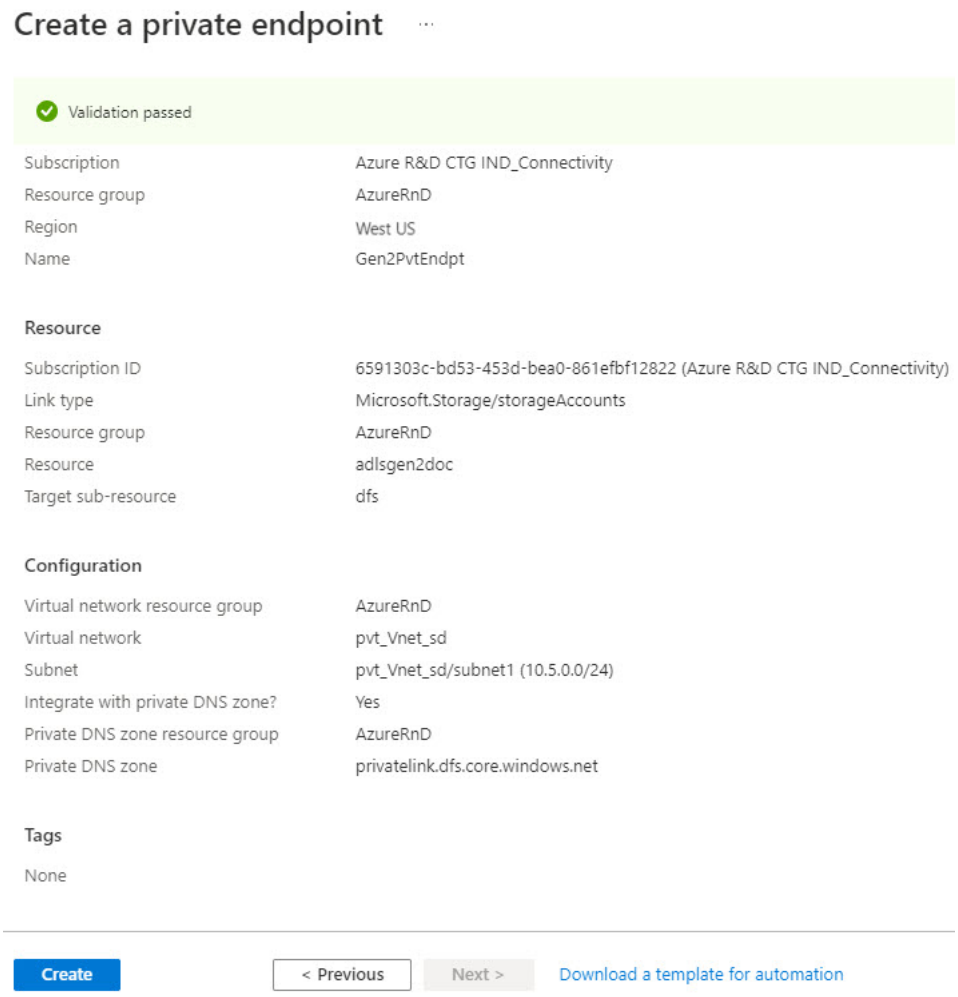


7. On the **Configuration** tab, select the virtual network and subnet that you created, and then click **Next : Tags**.



8. Optional. On the **Tags** tab, create tags to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups, and then click **Next : Review + Create**.

9. On the **Review + Create** tab, verify the details, and then click **Create**.

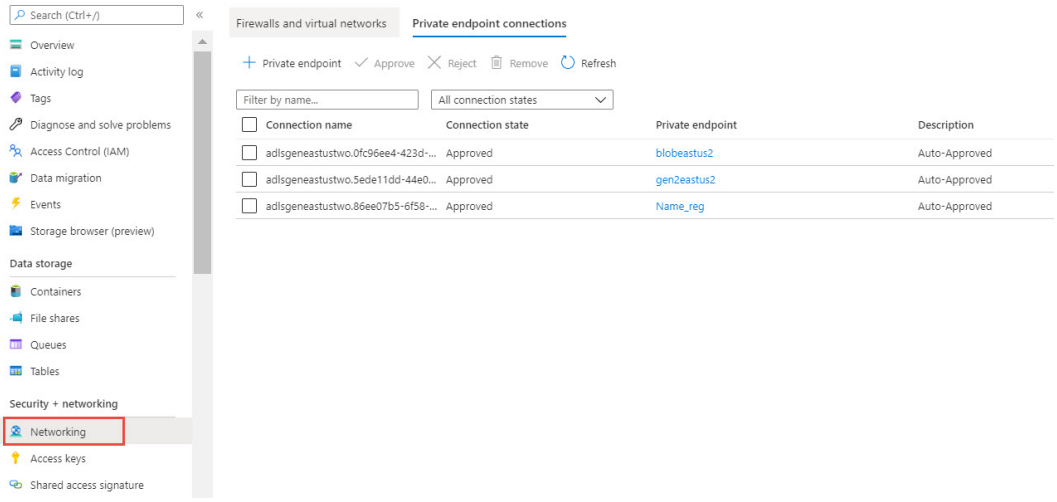


Create a virtual network link

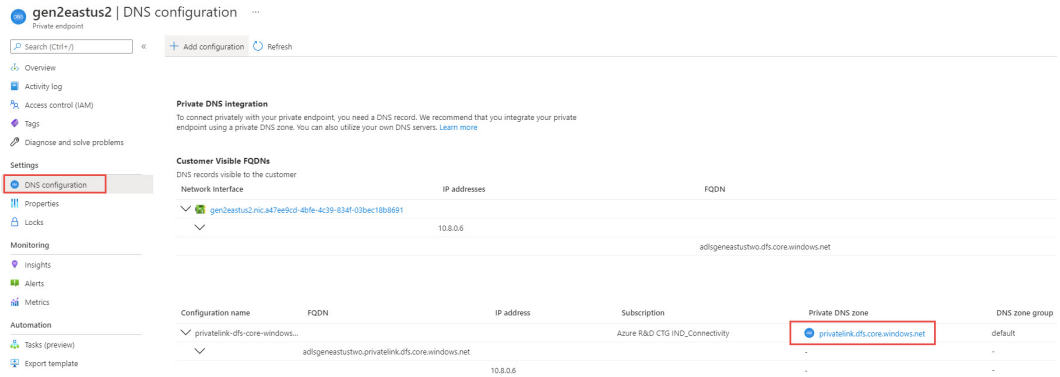
After you create a private endpoint, you must create a virtual network link to link the virtual network to the private DNS zone of the private endpoint. Once linked, virtual machines hosted in that virtual network can access the private DNS zone.

Perform the following steps to create a virtual network link for the private endpoint:

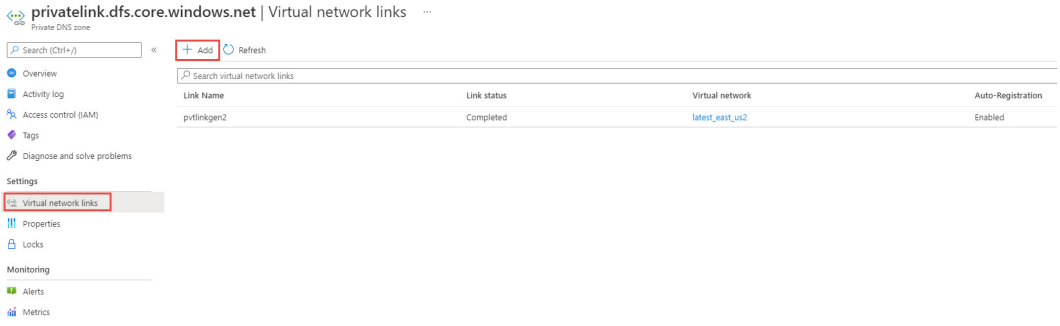
1. Go to the **Networking** tab of the storage account for which you have created the private endpoint.
2. On the **Private endpoint connections** tab, click the name of the private endpoint.



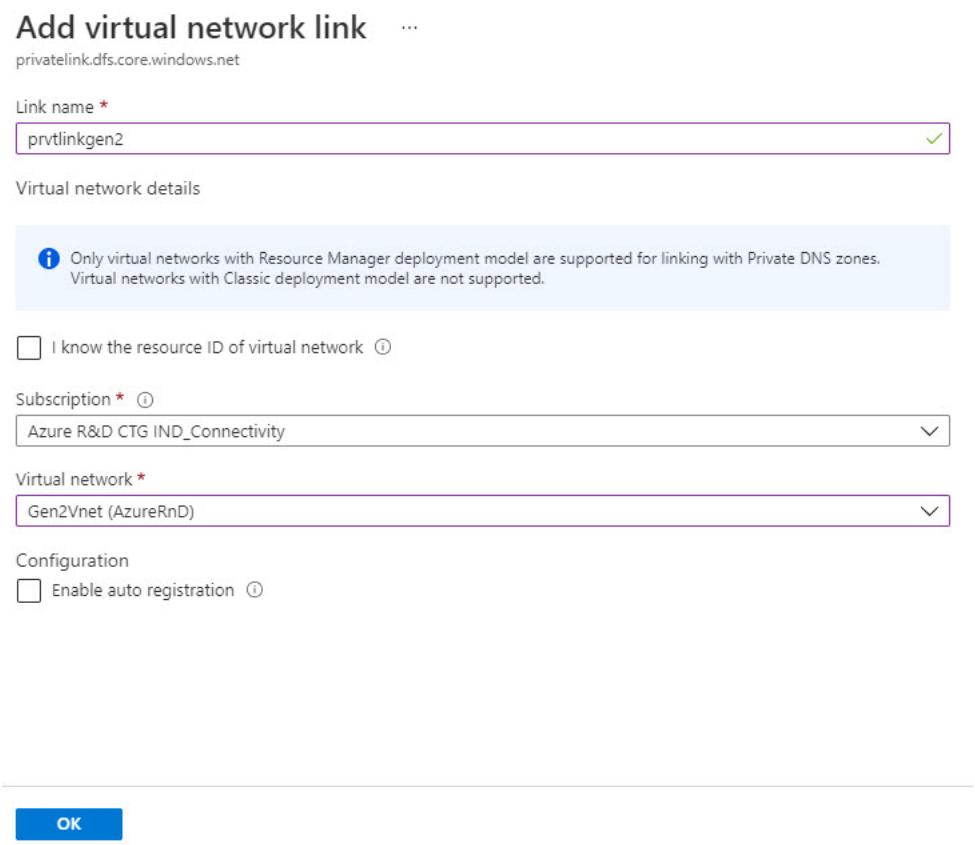
3. Click **DNS configuration**.



4. On the **DNS configuration** page, click the Private DNS zone for the private endpoint.
5. Click **Virtual network links**, and then click **Add**.



6. In the **Link name** field, enter a name for the virtual network link.



7. Select the **Subscription** and the **Virtual network** you want to link with.
8. Click **OK**.

Verify the Azure private endpoint configuration

After you configure the private endpoint, you can verify if the agent uses the Azure private endpoint to connect to the Azure storage.

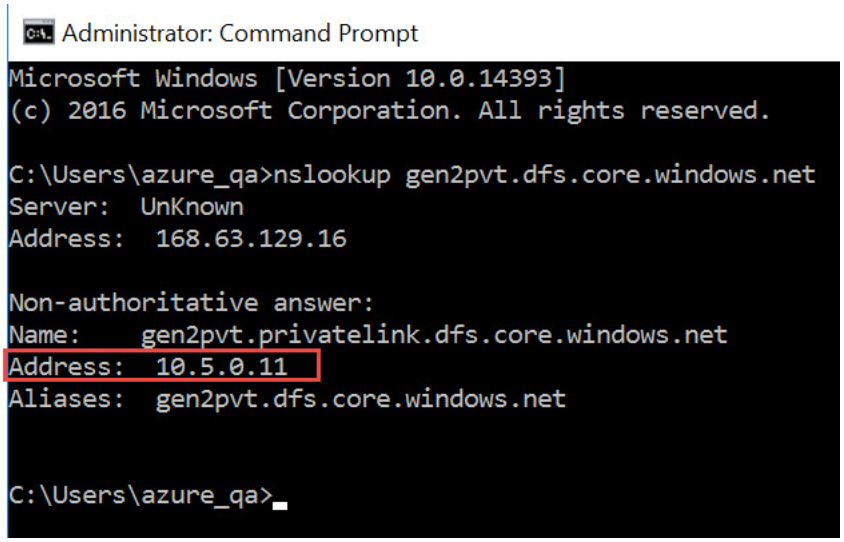
Perform the following steps to verify the Azure private endpoint configuration:

1. Log in to the virtual machine that you created.

2. Open the command prompt and enter the command in the following format:
   `nslookup <storage account name>.dfs.core.windows.net`

   For example, `nslookup gen2pvt.dfs.core.windows.net`

   If the agent uses the private endpoint to connect to the Azure storage, the command prompt shows the IP address of the private endpoint.



Author

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